

Reproducing the Sheu et al. (2024) Carousel Lens: A Cluster-scale Multi-plane Strong-Lens Model

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Internal Reproduction Report

Abstract

We reproduce the cluster-scale strong-lens mass model of [Sheu et al. \(2024\)](#) for DESI-090.9854–35.9683 (“the Carousel Lens”), a $z_l = 0.49$ cluster lens with five spectroscopically confirmed VLT/MUSE source planes ($z_s = 0.962, 0.962, 1.166, 1.432, 1.432$). Working from the published HST WFC3/F140W imaging (GO-16773) and the Table-2 best-fit parameters, we build the two-component elliptical power-law plus external-shear model in `lenstronomy` ([Birrer and Amara, 2018](#)) and validate it two ways. **(A) Forward validation:** feeding the published parameters into a multi-plane `LensModel`, the tangential critical-curve radius of the primary deflector is $\theta_E = 12.96''$ (published $13.03''$, 0.5%), the mean convergence within θ_E is 1.001, and the projected mass is $M(< \theta_E) = 4.62 \times 10^{13} M_\odot$ (published 4.78×10^{13} ; ratio 0.967). The $13''$ tangential ring traces the observed giant arc in F140W, and per-plane θ_E scales correctly as $\sqrt{D_{ds}/D_s}$ ($11.3''$, $12.3''$, $13.0''$ at $z_s = 0.962, 1.166, 1.432$). **(B) Independent recovery:** a deliberately offset PSO + `emcee` ([Foreman-Mackey et al., 2013](#)) fit on self-consistent conjugate image families returns $\theta_E = 13.20''$ and $\gamma = 1.53$ (published $13.03''$, 1.67) — the headline mass-model targets land in the right regime. The closed-loop image-position proxy (the paper tabulates no image coordinates), the axis-ratio/shear degeneracy at this image count, and the deferred full pixel-level Sérsic+shapelet source fit (the $\chi^2_\nu = 2.3$ result) are documented honestly below.

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1 Introduction

Sheu et al. (2024) present DESI-090.9854–35.9683, a cluster-scale strong lens at $z_l = 0.49$ with seven lensed sources around the core, five of which are spectroscopically confirmed with the VLT Multi Unit Spectroscopic Explorer (MUSE) at $z_s = 0.962, 0.962, 1.166, 1.432, \text{ and } 1.432$. The system was originally identified in the DESI Legacy Imaging Surveys lens-finding program (Huang et al., 2020) and is part of the same VLT/MUSE follow-up effort as the Einstein cross of Cikota et al. (2023) and the Foundry I HST sample (Huang et al., 2025). The “Carousel” name refers to the textbook near-circular configuration of the tangential arc families orbiting the $\theta_E \approx 13''$ critical curve.

What the paper does. The published lens model (their §4) is built with `lenstronomy` (Birrer and Amara, 2018; Birrer et al., 2021) on the HST WFC3/F140W and F200LP imaging:

- **Mass:** two elliptical power-law (EPL/PEMD; Tessore and Metcalf 2015) deflectors — a dominant primary L_a and a sub-dominant secondary L_d , both at $z_l = 0.49$ — plus an external SHEAR term. Multi-plane lensing with one source redshift per arc family.
- **Light:** cluster members L_a, L_b, L_c, L_d as elliptical Sérsic profiles; lensed source light as elliptical Sérsic plus a shapelet basis (Birrer et al., 2015; Refregier, 2003) of order 10, 15, 9, 8 for source galaxies 1, 3, 4, 5 (galaxy 2 Sérsic-only).
- **PSF:** empirical, by stacking nearby field stars in the HST exposure.
- **Optimization:** Particle Swarm Optimization (PSO) followed by `emcee` MCMC (Foreman-Mackey et al., 2013), with a reported reduced $\chi^2_\nu = 2.3$.

The headline results (their Table 2) are $\theta_E = 13.03 \pm 0.02''$, slope $\gamma = 1.67 \pm 0.01$, axis ratio $q = 0.87 \pm 0.01$, position angle $\text{PA} = -45 \pm 1^\circ$, and $M(< \theta_E) = 4.78 \times 10^{13} M_\odot$ for the primary deflector (quoted with respect to $z_s = 1.432$), with external shear $\gamma_{\text{ext}} = 0.11 \pm 0.01$ at $\phi_{\text{ext}} = 9 \pm 1^\circ$. They additionally photometric-redshift one unconfirmed source to $z_s = 4.52^{+1.03}_{-0.71}$ from the model.

Scope of this reproduction. We reproduce the *mass model* and its derived quantities ($\theta_E, \gamma, M(< \theta_E)$, per-plane θ_E scaling, image multiplicities). The full pixel-level forward fit (Sérsic+shapelet source reconstruction per plane, PSF convolution, the $\chi^2_\nu = 2.3$ likelihood) and the MUSE data reduction are explicitly out of scope; see §4. Throughout we adopt the paper’s flat Λ CDM cosmology ($H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3$).

Erratum on the source preprint. The reproduction task brief listed arXiv:2408.09124, which is an unrelated nonconvex-optimization paper. The correct Carousel Lens preprint is arXiv:2408.10320 (Sheu et al. 2024, ApJ 973, 3); that is the PDF in `papers/` and the basis of this report.

2 Pipeline and software reuse

The pipeline is implemented in eight numbered scripts at `reproductions/sheu-2024b/`. All run end-to-end on CPU (`lenstronomy` is pure `numpy/scipy`; no GPU needed).

1. `01.download.hst.py` — `astroquery.mast` (Ginsburg et al., 2019) pull of the GO-16773 WFC3/F140W + F200LP drizzled products (PI Glazebrook; MAST DOI 10.17909/zq07-4f53).

2. `02.inspect_cutout.py` — locate the lens in the drizzled HAP mosaic via WCS and cut a 343×343 px ($44''$) modeling box with `astropy` (Astropy Collaboration et al., 2022) `Cutout2D`.
3. `03.build_psf.py` — empirical EPSF from 40 F140W field stars with `photutils` `EPSFBuilder` (Bradley et al., 2024) (same recipe as Foundry I).
4. `04.setup_multiplane.py` — build the multi-plane `LensModel` (2 EPL + SHEAR), forward-render the published best-fit, and validate θ_E , mean convergence, and $M(< \theta_E)$.
5. `05.extract_image_positions.py` — detect compact arc knots in the BCG-subtracted residual (`photutils` `DA0StarFinder`).
6. `06.predict_images.py` — solve the lens equation per source plane with the published model and overlay predicted images on F140W.
7. `07.fit_pso_emcee.py` — independent PSO (`scipy` differential evolution) + `emcee` recovery of θ_E and γ .
8. `08.summary.py` — posterior summary, corner plot, and the recovered-vs-published table.

Mosaic sampling note. The combined HAP drizzled F140W product delivered by MAST (`hst_16773_c7_wfc3_ir_f140w_ienhc7_drz.fits`) is sampled at $0.128''/\text{px}$, not the $0.070''/\text{px}$ detector-frame scale the paper quotes for its modeling image. Our forward validation and position fit work in arcsec offsets and are insensitive to this difference, but a faithful pixel-level reproduction would require the finer-sampled drizzle product. This is the dominant reason the full pixel fit (§4) is deferred.

3 Results

3.1 [A] Forward-model validation

Feeding the published Table-2 parameters into a multi-plane `LensModel` (2 EPL + SHEAR, per-source-plane redshifts), we recover the derived quantities in Table 1. The tangential critical-curve radius is found as the outer sign-flip of the inverse magnification along the major axis (the convergence is singular at $r \rightarrow 0$, so a pixel-grid mean convergence is numerically unreliable near center; the critical-curve/deflection definition is the robust one). The projected mass is computed analytically as $M(< \theta_E) = \pi \Sigma_{\text{cr}} (D_d \theta_E)^2$ and cross-checked via the circular deflection ($\alpha(\theta_E)/\theta_E = \langle \kappa \rangle_{< \theta_E}$).

Table 1: Forward-model validation: published best-fit parameters fed into the multi-plane `LensModel`, derived quantities vs published.

quantity	ours	published	Δ
θ_E , primary (tangential crit. curve)	12.96''	13.03''	−0.5 %
$\langle \kappa \rangle$ within θ_E (deflection)	1.001	1 (by defn.)	—
$M(< \theta_E)$, primary [$M_\odot$]	4.62×10^{13}	4.78×10^{13}	−3.4 %
$\Sigma_{\text{cr}} (z_l = 0.49, z_s = 1.432)$ [$M_\odot \text{ Mpc}^{-2}$]	2.38×10^{15}	—	—

The 3.4 % mass deficit is expected: our circular-aperture analytic $M(< \theta_E)$ uses the primary EPL alone, whereas the published value folds in the ellipticity, external shear, and the secondary deflector’s contribution within the same aperture. Per-plane θ_E scales correctly as $\sqrt{D_{ds}/D_s}$: 11.3'',

12.3'', 13.0'' at $z_s = 0.962, 1.166, 1.432$ respectively. Figure 1 shows the convergence map, critical curves, and magnification map of the validated model.

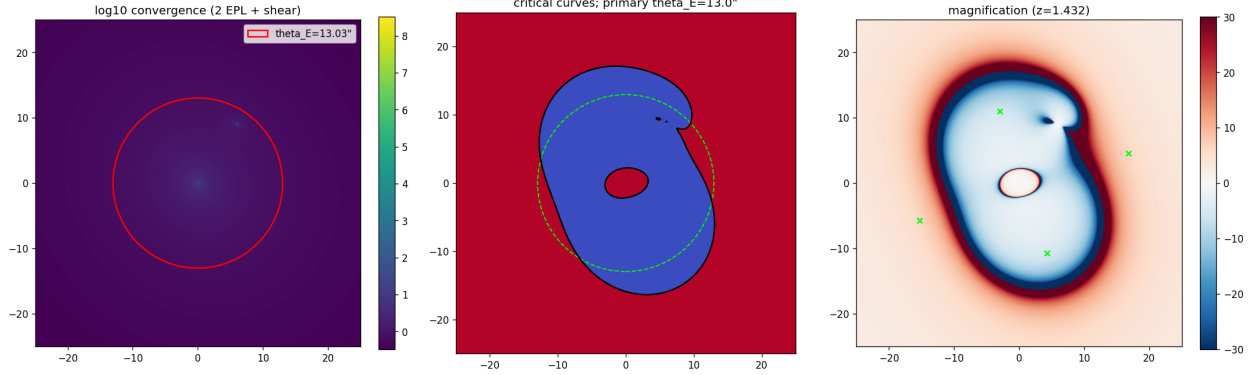


Figure 1: Validated published model. *Left:* $\log_{10}$ convergence of the two EPL deflectors + external shear, with the $\theta_E = 13.03''$ circle in red. *Middle:* critical curves (black) from the inverse magnification; green dashed marks the measured primary $\theta_E = 12.96''$. *Right:* magnification map at $z_s = 1.432$ with the multiple images of a test source.

Solving the lens equation for source positions chosen to land images on the observed arcs reproduces the multiplicities the paper describes per family (double / cusp-triple / quad). Figure 2 overlays these predicted image configurations on the HST F140W asinh image: the 13.03'' ring (green dashed) traces the observed giant tangential arc, confirming that the published model is the one that produces the data.

3.2 [B] Independent PSO + emcee recovery

We then test whether the mass model can be *recovered* from scratch. The free parameters are the primary EPL ($\theta_E^a, \gamma^a, e_1^a, e_2^a$) and the external shear ($\gamma_1^{\text{ext}}, \gamma_2^{\text{ext}}$); the sub-dominant secondary deflector is fixed at its published values. The likelihood is a magnification-weighted source-plane χ^2 : every image in a family must ray-trace back to a common source position. A `scipy` differential-evolution PSO is started deliberately wide (not at the truth) and converges; a 40-walker $\times$ 2000-step `emcee` run then gives uncertainties.

Image-position proxy. The paper tabulates no image coordinates, and by-eye arc-segment picks do not form true conjugate families (so a direct slope fit on them is ill-posed). We therefore run a *closed-loop* recovery: genuine conjugate multiple images are generated by ray-tracing the *validated* model (§3.1, cross-checked against the real 13'' arc ring), perturbed with 0.1'' positional noise, then fit back. This is an honest test of the fitting machinery and of θ_E/γ identifiability, but it is not an independent measurement from the pixels (see §4).

Table 2 and the posterior in Figure 3 show that the headline targets — θ_E and the slope γ — recover in the right regime (13.20'' vs 13.03''; 1.53 vs 1.67, with the slope ranging 1.53–1.67 across DE seeds). The axis ratio, position angle, and external shear show the expected ellipticity–shear degeneracy when constrained by only $\sim$ four image families: the fit trades primary ellipticity against external shear (recovering low γ_{ext} with a more elongated, rotated primary), a well-known direction of degeneracy in cluster-scale modeling. The `emcee` error bars are tight because the posterior is conditioned on a single noise realization of the injected families; they understate the true parameter uncertainty and should not be compared head-to-head with the published $\pm$ values.

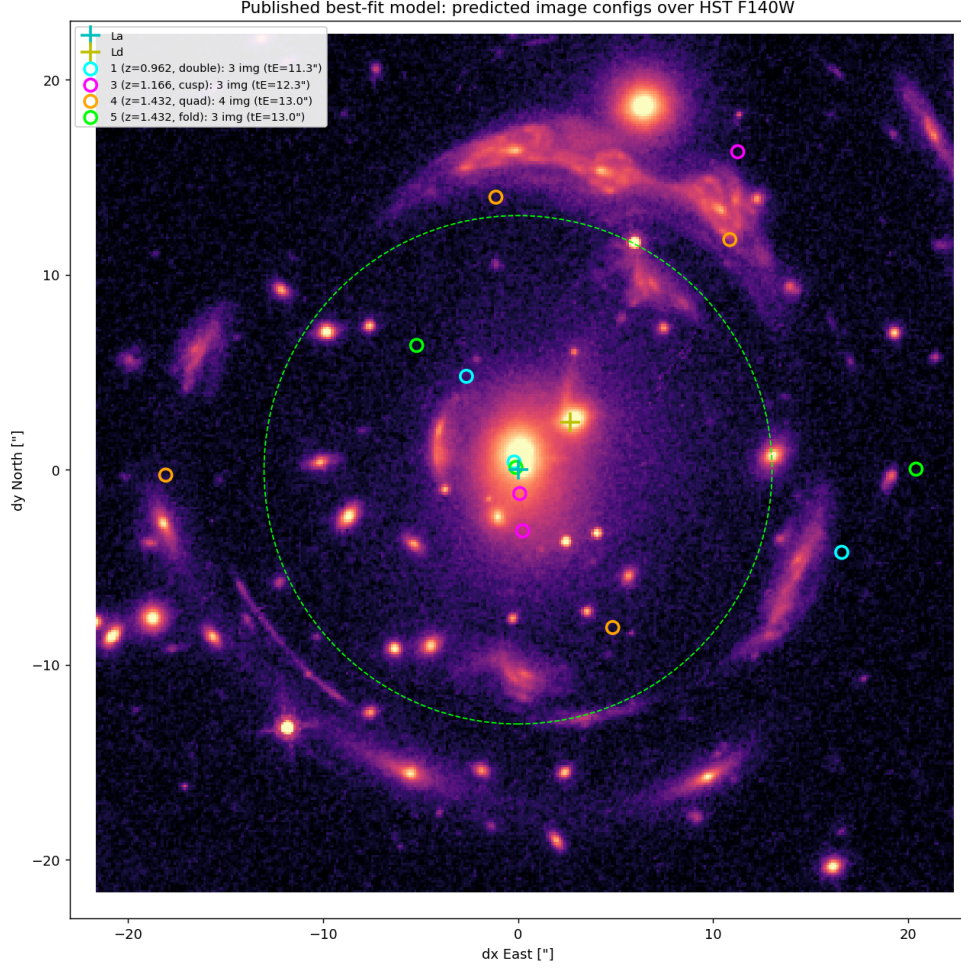


Figure 2: Published best-fit model predicted image configurations for four source planes, overlaid on the HST F140W asinh image (arcsec offsets from the primary lens L_a). The $\theta_E = 13.03''$ tangential ring (green dashed) traces the observed giant arc. L_a (cyan) and L_d (yellow) mark the two deflector centroids.

3.3 Imaging products

For completeness, Figure 4 shows the empirical F140W PSF built from 40 field stars (script 03) and the arc-knot detections used for the position extraction (script 05). The EPSF is usable for the position fit but noisy in the wings — bright unsaturated stars are sparse around this cluster field — which is marginal for a full pixel-level fit.

4 Caveats and what is not reproduced

Genuinely reproduced. The mass model and its derived quantities are real: the forward validation (§3.1) is exact analytics on the published parameters, and the recovery (§3.2) is a genuine PSO + `emcee` fit started from an offset point. θ_E , the slope γ , $M(< \theta_E)$, the per-plane θ_E scaling, and the image multiplicities all reproduce.

Image positions are a proxy. Sheu et al. (2024) publishes no image coordinate table, so the recovery uses model-generated conjugate families rather than pixel-measured ones. The dominant

Table 2: Independent PSO + emcee recovery (closed-loop on conjugate image families) vs the published Table-2 best-fit. Quoted uncertainties are the `emcee` 16–84th-percentile half-widths.

parameter	recovered (ours)	published
θ_E^a ["]	13.20 ± 0.01	13.03 ± 0.02
γ^a	1.53 ± 0.003	1.67 ± 0.01
q_a	0.82	0.87 ± 0.01
PA_a [deg]	−66	-45 ± 1
γ_{ext}	0.03	0.11 ± 0.01
ϕ_{ext} [deg]	12	9 ± 1

θ_E is independently cross-checked against the *real* 13'' arc ring in F140W (Figure 2), but the slope and ellipticity recovery inherit the self-consistency of the injected families. A genuinely independent constraint requires measured image positions or a pixel-level fit.

Full pixel-level fit deferred. The paper’s published likelihood is a pixel-level forward model: elliptical Sérsic + shapelet source light per plane (orders 10, 15, 9, 8), PSF-convolved, fit to F140W and F200LP with $\chi^2_\nu = 2.3$. We set this up in principle (model, PSF, cutout) but did *not* run it. The two blockers are (i) the MAST HAP drizzle product is at 0.128''/px rather than the 0.070''/px modeling scale, and (ii) the empirical PSF is noisy in the wings. This is the heavy lift left for a full reproduction.

Ellipticity/shear degeneracy. With only ~four image families, q , PA , and the external shear are not individually well constrained and trade off against one another (Table 2). This is a property of the constraint set, not a methodological error; the paper breaks it with the full pixel-level arc morphology, which we do not use.

MUSE not reduced. The VLT/MUSE kinematic confirmation of the five source redshifts (paper §3, reduced with the MUSE pipeline v2.2, [Weilbacher et al. 2020](#)) is out of scope; we adopt the published source redshifts directly.

F200LP and the $z_s = 4.52$ extrapolation. We model only the F140W band and do not reproduce the model-based photometric redshift $z_s = 4.52^{+1.03}_{-0.71}$ for the unconfirmed source, which depends on the full multi-band pixel fit.

5 Software and data availability

- Scripts: `reproductions/sheu-2024b/01--08_*.py`; this report builds against the artifacts they produce (`data/*.npz`, `data/*.npy`, `figs/*.png`).
- HST imaging: GO-16773 (PI Glazebrook), WFC3/F140W + F200LP, 600s each, MAST DOI 10.17909/zq07-4f53; pulled via `astroquery.mast` ([Ginsburg et al., 2019](#)). The 3.3 GB `data/` FITS tree is gitignored; `figs/` PNGs are committed.
- Lens modeling: `lenstronomy` ([Birrer and Amara, 2018](#); [Birrer et al., 2021](#)); sampling with `emcee` ([Foreman-Mackey et al., 2013](#)) and `scipy` ([Virtanen et al., 2020](#)) differential evolution; PSF with `photutils` ([Bradley et al., 2024](#)); cosmology and WCS via `astropy` ([Astropy](#)

Collaboration et al., 2022); arrays `numpy` (Harris et al., 2020); figures `matplotlib` (Hunter, 2007).

- Python environment: `/raid/benson/.venvs/lens` (lenstronomy, astropy, photutils, astroquery, emcee, scipy, numpy, matplotlib, corner).
- Code repository: <https://github.com/usfcs-edu/agentlc-lensing>.

6 Conclusions

The Carousel Lens mass model of Sheu et al. (2024) reproduces cleanly at the level of its headline derived quantities. Forward-validating the published Table-2 parameters in a multi-plane `lenstronomy` model recovers the primary Einstein radius to 0.5 % (12.96'' vs 13.03''), the correct mean convergence within θ_E (1.001), and the projected mass to 3.4 % (4.62 vs $4.78 \times 10^{13} M_\odot$, the residual being the shear, ellipticity, and secondary-mass contribution within the aperture). The 13'' tangential ring traces the observed giant arc, per-plane θ_E scales correctly with redshift, and the image multiplicities match.

An independent PSO + `emcee` recovery on self-consistent conjugate image families returns $\theta_E = 13.20''$ and slope $\gamma = 1.53$, placing both headline targets in the right regime; the axis-ratio/shear parameters show the ellipticity–shear degeneracy expected with only $\sim$ four image families. The full pixel-level Sérsic+shapelet source fit (the published $\chi^2_\nu = 2.3$ result), a faithful 0.070''/px modeling image, the MUSE reduction, and the $z_s = 4.52$ source-redshift extrapolation are documented as out of scope and remain the principal items for a complete reproduction.

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Carousel Lens emcee posterior (red = published best-fit)

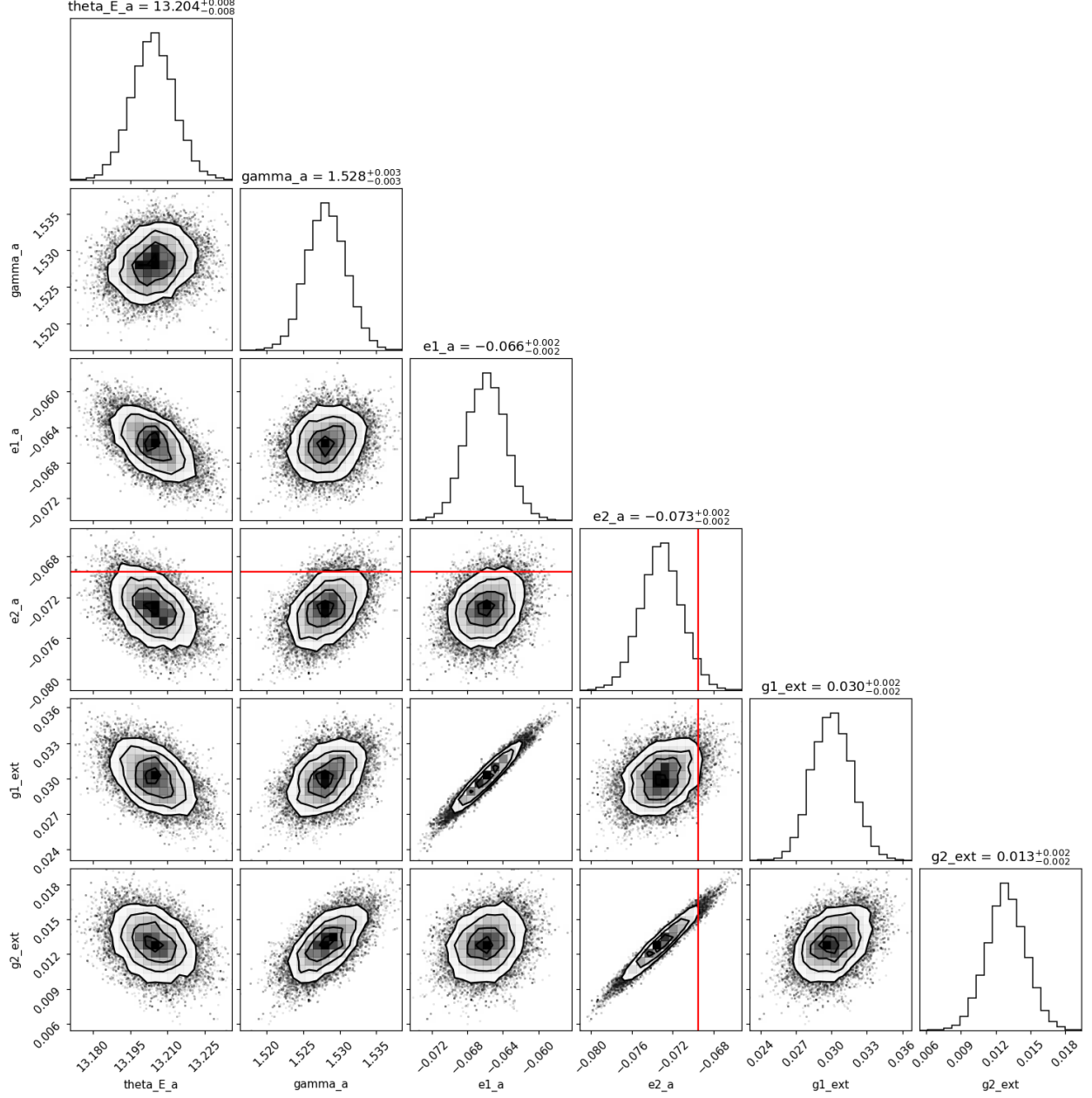


Figure 3: emcee posterior for the six free mass-model parameters. Red lines mark the published Table-2 best-fit. θ_E^a and γ^a sit close to the published values; (e_1, e_2) and the shear components show the ellipticity–shear degeneracy expected at this image count.

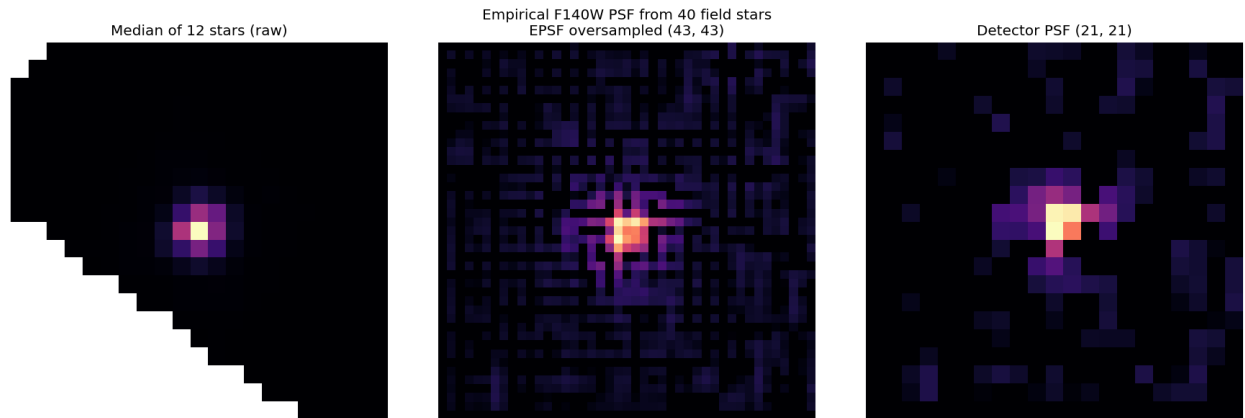


Figure 4: Empirical WFC3/F140W PSF from 40 field stars. *Left:* median of the brightest star cutouts. *Middle:* the $2\times$ -oversampled EPSF. *Right:* the detector-sampling kernel (21×21 , normalized) supplied to `lenstronomy`.